PH 724 Data Analysis & Scientific Communication Incorporating Use of Generative AI

Part 6: Statistical Methods and Reporting in Clinical Research

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[P#3] Barkan (2015)

Central message

Statistical validity starts with the research question, study design, outcome definition, and data structure. Statistical tests cannot repair fundamental design problems.

Main points

- **Measurement scales:** Identify whether variables are nominal, ordinal, or numerical before choosing summaries and analyses.
- **Association versus causation:** Correlation and regression quantify relationships; neither alone proves a causal effect.
- **Match models to outcomes:** Linear regression is commonly used for continuous outcomes, logistic regression for binary outcomes, and survival methods such as Cox regression for time-to-event data with censoring.
- **Account for dependence:** Independent groups, paired before–after observations, and repeated measurements require different analytical approaches.
- **Inference and uncertainty:** Interpret P values alongside effect estimates, confidence intervals, assumptions, and clinical relevance.
- **Sample size and power:** Plan the study around a meaningful target effect, expected variability or event rate, desired power, and an error-rate threshold.
- **Nonsignificant results:** Failure to reject a null hypothesis does not establish equivalence or absence of an effect.
- **Pilot studies:** Pilot work can assess feasibility, refine measurement procedures, and inform assumptions for a larger study.

Practical implication

Choose an analysis strategy *before* collecting data where possible, and align every statistical method with the question and study design.

[P#4] Habibzadeh (2017)

Central message

Good reporting communicates not only whether an association or effect is statistically significant, but also its magnitude, precision, uncertainty, and practical importance.

Main points

- **Numbers and percentages:** Avoid unjustified decimal precision. Give counts and denominators alongside percentages, especially for small samples.
- **Descriptive statistics:** Use mean and standard deviation (SD) for reasonably symmetric distributions; median and interquartile range (IQR) are often better for skewed distributions. SD describes variability in observations; standard error describes uncertainty in an estimated mean.
- **Confidence intervals:** Present effect estimates with confidence intervals, commonly 95%, to convey precision and uncertainty.
- **P values:** Report exact P values when appropriate (e.g., $P = 0.032$); do not write $P = 0.000$. Statistical significance is not synonymous with clinical importance.
- **Multiple comparisons:** Repeated testing raises the risk of false positives. Use appropriate multiplicity adjustments when warranted, and distinguish prespecified from exploratory analyses. The false discovery rate (FDR) control and Bonferroni adjustments are common approaches.
- **Risk measures:** Report relative measures (risk ratios, odds ratios, hazard ratios) with uncertainty intervals and provide absolute risks or risk differences for context.
- **Diagnostic accuracy:** Report sensitivity, specificity, predictive values, likelihood ratios, and ROC/AUC measures as relevant; specify cutoffs and note that predictive values depend on prevalence.
- **Methods transparency:** Describe analytical choices, assumptions, sample-size planning, missing data, outliers, and software sufficiently for critical appraisal.

Practical implication

A reader should be able to reconstruct what was measured, how results were summarized, how uncertain the estimates are, and whether observed effects matter in practice.

Bottom Line

Barkan (2015) focuses on obtaining defensible results through sound design and analysis; Habibzadeh (2017) focuses on communicating those results accurately and meaningfully. Together, they offer a practical framework from study planning through publication.

References

1. Barkan H. (2015) Statistics in clinical research: Important considerations. *Annals of Cardiac Anaesthesia.* 18:74–82. https://doi.org/10.4103/0971-9784.148325
2. Habibzadeh F. (2017) How to report the results of public health research. *Journal of Public Health and Emergency* 1:1–8. https://doi.org/10.21037/jphe.2017.12.02